Nithisha Venkatesh

nithishaleni1806@gmail.com | linkedin.com/in/nithisha-v | github.com/NithishaVenkatesh | +91 6379 074 239

SUMMARY

Applied AI Engineer specializing in agentic systems and production-grade RAG, with a backend-first focus on FastAPI, data pipelines, and LLM orchestration (LangGraph, OpenAI Agentic SDK). Built a multi-agent tutoring platform(Winner, Microsoft Skill Fest Student Award) with a hardened safety layer and citation-grounded retrieval, and co-built NextHire AI, a SaaS agentic developer-evaluation product.

PRODUCTS

NextHire AI – Dev Evaluation Platform for Agentic Era — **nexthire-ai.tech**

- Co-built a **SaaS platform** that evaluates developers by assigning a real, role-specific production bug to fix in their own IDE, capturing full session trajectories (every prompt, edit, command, run) for **evidence-based scoring**, built on **Next.js and Convex.**
- Engineered a containerized (**Docker**) GitHub indexer with a **multi-gate filtering pipeline**, **Levenshtein-based** tech-stack extraction, paid-API exclusion, and **CI/CD YAML**-based local-runnability checks, persisting qualified repositories to **Neon Postgres** for SQL retrieval against any job description.
- Built an **evidence-grounded** agentic evaluation pipeline (**OpenAI Agentic SDK, Kimi API**) generating a per-competency scorecard across 11 clusters, with every score grounded in the candidate's actual session rather than pass/fail signals.
- Integrated **Groq inference** to power a low-latency AI assistant supporting recruiters throughout the hiring workflow.

PROJECTS

Athenaeum – Multi-Agent Learning System — **github.com/theCodeForgerHQ/msf-reasoning-agent**

- Co-architected a **red-team-hardened**, multi-agent tutoring platform (**FastAPI**, Next.js 16, Python 3.12) on **Microsoft Foundry**, covering learning-path curation, capacity-aware study planning, citation-grounded tutoring, and graded assessments.
- Designed a **3-node reasoning pipeline** (gate → router → answer) streaming **12 typed SSE** event types with a phase-by-phase trace exposing model, tier, latency, confidence, and grounding source - **making reasoning inspectable** rather than hidden in a chat window.
- Engineered a **5-layer defense-in-depth safety gate** (regex and deterministic heuristics handling 24 homoglyphs, Groq Prompt Guard 2, Azure FAST classifier, **Azure RAI filter**), validated across 13 adversarial batteries and 232 cases, achieving a **100% direct block-rate** and **96% converter-sweep block-rate** with zero over-refusal regression.
- Built hybrid **Foundry IQ grounding** (Azure AI Search agentic retrieval plus **IDF-gated lexical fallback**), Azure AI Evaluation scoring (groundedness, relevance, retrieval), and a re-dispatch reflection loop ensuring every cited claim is verifiable.
- Implemented a **4-tier model fallback** chain with per-provider circuit breakers, exponential-backoff retry, and a typed unsafe-vs-unreachable distinction so safety refusals never silently fail open.

Nucleus AI – Enterprise Knowledge Platform — **github.com/theCodeForgerHQ/Nucleus-AI**

- Co-architected a **multi-service RAG system** (FastAPI, Docker) ingesting structured knowledge from Confluence, delivering citable responses through reliability-first validation layers.
- Built a fault-tolerant ingestion pipeline with **staged indexing**, **webhook-driven updates**, retry logic, and **idempotent upserts**, preventing index corruption during iterative knowledge updates.
- Implemented **hybrid retrieval** (vector search + cross-encoder reranking) with **intent-aware filtering**, improving retrieval precision while **reducing redundant inference calls**.
- Instrumented stage-level latency decomposition (**~7s** local end-to-end) across retrieval, **reranking**, generation, and validation using **LangSmith** trace pipelines.

EXPERIENCE

Artificial Intelligence Engineer · iQube — **June 2024 – Present**

- **Designed, developed, and deployed scalable end-to-end AI systems** spanning machine learning, deep learning, and generative AI, building production-ready FastAPI-based platforms with RAG pipelines, vector databases, hybrid retrieval, cross-encoder reranking, LLM integration, and robust observability via latency tracing and structured logging.
- **Built AI-driven analytics and predictive solutions** by architecting end-to-end data pipelines for preprocessing, EDA, dimensionality reduction, and model optimization, delivering time-series forecasting, risk classification, and financial analytics models on large-scale datasets for real-world decision-making.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, SQL.
AI & LLM Systems: OpenAI Agentic SDK, Claude Agentic SDK, RAG Architecture, Hybrid Retrieval, Vector Search, Embeddings, LLM Evaluation, Guardrails, NLI-based Validation, LangGraph.
AI Tooling & Models: Cohere, Pinecone, Milvus, LangSmith, PromptLayer, HuggingFace, TensorFlow, PyTorch.
Backend: FastAPI, REST APIs, PostgreSQL(Neon), Drizzle ORM, ClickHouse, Caching.
Infrastructure & DevOps: Docker, Vercel, n8n, Observability (Metabase).

ACHIEVEMENTS

- **Winner**, Microsoft Agent League Hackathon (Student Award) - Project Athenaeum - 2026
- **Achiever's Award**, National Innovation Challenge for Drone Application & Research (NIDAR)- 2026
- **Finalist**, Zoho Creator Hackathon - 2025

EDUCATION

B.Tech, Artificial Intelligence & Data Science - Kumaraguru College of Technology - CGPA: 8.2